

10 Time Reversal Symmetry in Quantum Mechanics

In classical mechanics, time reversal has the following effect on physical quantities:

$$t(\text{time}) \rightarrow -t, x(\text{position}) \rightarrow x, p(\text{momentum}) \rightarrow -p, E(\text{electric field}) \rightarrow E, B(\text{magnetic field}) \rightarrow -B \quad (1)$$

In quantum mechanics, we would expect similar relations to hold under time reversal. However, there is a problem. Consider the canonical commutation relation between position and momentum

$$[x, p] = i\hbar \quad (2)$$

If $x \rightarrow x$, $p \rightarrow -p$ under time reversal, then this commutation relation no longer holds.

How to solve this problem?

It was realized by Wigner that in quantum mechanics, time reversal has to be defined in a very special way different from all other symmetries. Time reversal operator is anti-unitary: it maps i to $-i$. If this is the case, then the canonical commutation relation between x and p will remain invariant under time reversal.

The action of time reversal on a wave function then contains two parts: first take complex conjugation in a certain basis, then apply a linear transformation in this basis. That is, we can write

$$\mathcal{T} = UK \quad (3)$$

where K denotes complex conjugation and U denotes some unitary transformation.

Then time reversal acts on operators as

$$\mathcal{T}O\mathcal{T}^{-1} = UKOKU^\dagger = UO^*U^\dagger \quad (4)$$

That is, the action of time reversal on operators contains two parts: first take complex conjugation of the operator written in certain basis, then conjugate the operator by some unitary transformation in this basis.

Let's see what this anti-unitary property of time reversal implies.

0. Time reversal does not commute with complex numbers.

$$\mathcal{T}c\mathcal{T}^{-1} = c^* \quad (5)$$

1. Time reversal is not a linear operator on wave functions.

$$\mathcal{T}(\alpha_1\psi_1 + \alpha_2\psi_2) = \alpha_1^*\mathcal{T}(\psi_1) + \alpha_2^*\mathcal{T}(\psi_2) \quad (6)$$

Instead it is said to be anti-linear.

Are anti-linear operators allowed as symmetry operators in quantum mechanics? The answer is yes, because:

2. Under time reversal, the absolute value of wave function inner product remains invariant.

To see this, notice that

$$\mathcal{T}\psi_1 = U\psi_1^*, \mathcal{T}\psi_2 = U\psi_2^* \quad (7)$$

Therefore,

$$\langle \mathcal{T}\psi_1 | \mathcal{T}\psi_2 \rangle = \langle \psi_1^* | U^\dagger U | \psi_2^* \rangle = \langle \psi_1 | \psi_2 \rangle^* \quad (8)$$

The inner product between any two wave functions does not remain invariant, but that is not a problem because the absolute value of the inner product is and only the absolute value is experimentally measurable. In fact, Wigner showed that in order for the absolute value of inner product be invariant, the symmetry operation must either be unitary or anti-unitary.

3. Representations of time reversal symmetry.

Time reversal, as its name suggests, forms a C_2 symmetry group, $\mathcal{T}^2 = 1$. However, the representations of time reversal symmetry are very different from that of usual C_2 symmetry groups, which have unitary representations.

Let's first look for one dimensional representations of time reversal symmetry. That is, we are looking for quantum states which transform under time reversal as

$$\mathcal{T}|\psi\rangle = e^{i\theta}|\psi\rangle \quad (9)$$

Applying time reversal again, we get

$$\mathcal{T}^2|\psi\rangle = e^{-i\theta}(\mathcal{T}|\psi\rangle) = e^{-i\theta}e^{i\theta}|\psi\rangle = |\psi\rangle \quad (10)$$

Therefore, the 'quantum number' of time reversal symmetry $e^{i\theta}$ can be any complex number of absolute value 1. It may seem that time reversal has an infinite number of one dimensional representations, but in fact they are all equivalent to each other. If we apply a basis transformation $|\psi'\rangle = e^{i\alpha}|\psi\rangle$, then

$$\mathcal{T}|\psi'\rangle = e^{-i\alpha}e^{i\theta}|\psi\rangle = e^{-i2\alpha}e^{i\theta}|\psi'\rangle \quad (11)$$

By tuning α , we can change from one θ to any other θ' . Therefore, in this sense, time reversal has only one (trivial) one dimensional representation. In fact, this is the only **linear** representation of time reversal.

On the other hand, time reversal as an interesting **projective** representation. Consider the action of

$$\mathcal{T} = 2i\sigma_y K = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} K \quad (12)$$

on the Hilbert space of a spin 1/2. $\mathcal{T}$ applies the expected operation to all spin operators

$$\mathcal{T}\sigma_x\mathcal{T}^{-1} = -\sigma_x, \mathcal{T}\sigma_y\mathcal{T}^{-1} = -\sigma_y, \mathcal{T}\sigma_z\mathcal{T}^{-1} = -\sigma_z \quad (13)$$

That is, it reverses the direction of spin – a magnetic dipole moment. However, this representation is not linear

$$\mathcal{T}^2 = -1 \quad (14)$$

It is a projective representation of time reversal and more interestingly it has to be at least two dimension (homework). As fermions all have spin 1/2, we say that fermions transform under time reversal as $\mathcal{T}^2 = -1$.

11 Group Theory and Standard Model

Group theory played a big role in the development of the Standard model, which explains the origin of all fundamental particles we see in nature. In order to understand how that works, we need to learn about a new Lie group: $SU(3)$.

11.1 $SU(3)$ and more about Lie groups

$SU(3)$ is the group of special ($\det U = 1$) unitary ($UU^\dagger = I$) matrices of dimension three. What are the generators of $SU(3)$? If we want three dimensional matrices X such that $U = e^{i\theta X}$ is unitary (eigenvalues of absolute value 1), then X need to be Hermitian (real eigenvalue). Moreover, if U has determinant 1, X has to be traceless. Therefore, the generators of $SU(3)$ are the set of traceless Hermitian matrices of dimension 3. Let's count how many independent parameters we need to characterize this set of matrices (what is the dimension of the Lie algebra). 3×3 complex matrices contains 18 real parameters. If it is to be Hermitian, then the number of parameters reduces by a half to 9. If we further impose traceless-ness, then the number of parameter reduces to 8. Therefore, the generator of $SU(3)$ forms an 8 dimensional vector space.

We can choose a basis for this eight dimensional vector space as

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (15)$$

$$\lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \quad (16)$$

They are called the **Gell-Mann matrices**. Among them λ_3 and λ_8 are diagonal and the other six correspond to the off-diagonal part of the Hermitian matrices. If we recall the structure of the Lie algebra of $SU(2)$, λ_3 and λ_8 are similar to J_z and others are similar to J_x and J_y .

Note that λ_1, λ_2 and λ_3 are exactly the J_x, J_y and J_z operators acting on the first two dimensions. Therefore, they form a $su(2)$ sub-algebra and generate an $SU(2)$ subgroup of $SU(3)$.

By convention, we define

$$T_a = \frac{1}{2} \lambda_a \quad (17)$$

Such that

$$\text{Tr}(T_a T_b) = \frac{1}{2} \delta_{ab} \quad (18)$$

From the T_a 's we can determine the commutator of the Lie algebra

$$[T_a, T_b] = i f_{abc} T_c \quad (19)$$

where f_{abc} is called the **structure constant** of the group and completely antisymmetric with respect to the exchange of any two indices. Recall for $SU(2)$, the structure constant was ϵ_{abc} which has a similar antisymmetric property.

The three dimensional special unitary matrices provide only one possible representation of $SU(3)$. This is called the **fundamental representation**. How to find the other representations? We can proceed in a similar way as $SU(2)$. Remember that for $SU(2)$, what we did was

1. find the Casimir operator which commute with the whole algebra and use its eigenvalue to label different representations
2. within each representation, use the eigenstates of J_z to label different basis states
3. define the raising and lowering operators $J_{\pm}$ to map from one basis state to another and determine the largest and smallest J_z eigenvalue given j .

Let's try to do something similar for $SU(3)$. First, the $su(3)$ algebra has two Casimir operators

$$C_1 = \sum_{i=1}^8 T_i^2, C_2 = \sum_{ijk} d_{ijk} T_i T_j T_k \quad (20)$$

where d_{ijk} can be obtained from the anti-commutation relation for the generators

$$\{T_i, T_j\} = T_i T_j + T_j T_i = \frac{1}{3} \delta_{ij} + d_{ijk} T_k \quad (21)$$

Direct calculation shows that $C_1 = \frac{4}{3}$, $C_2 = \frac{10}{9}$ for the fundamental representation. In fact, the value for C_1 and C_2 can be obtained from two integers p and q .

$$C_1 = \frac{1}{3}(p^2 + q^3 + 3p + 3q + pq), C_2 = \frac{1}{18}(p - q)(3 + p + 2q)(3 + q + 2p) \quad (22)$$

Therefore, instead of using C_1 and C_2 to label representations, we can use p and q and denote each irrep as $D(p, q)$. The fundamental representation is then labeled as $D(1, 0)$.

Now we want to find a set of basis states for each irrep. Similar to $SU(2)$, we can use the eigenstates of T_3 as basis states. In fact, because T_8 commute with T_3 , we can use their common eigenstates as basis states of the irrep. There are no other independent generator which commutes with both. Of course, we can make other choices of generators to define basis states, but it can be shown that different choices are all equivalent to each other and give the same results.

In a Lie algebra, the subset of commuting hermitian generators which is as large as possible is called the **Cartan subalgebra**. We can choose a basis for this subalgebra H_i such that

$$H_i = H_i^\dagger, [H_i, H_j] = 0, Tr(H_i H_j) = k_D \delta_{ij} (k_D \text{ is representation and normalization dependent}) \quad (23)$$

The H_i , $i = 1, \dots, m$, are called the **Cartan generators** and m is called the **rank** of the algebra.

The basis states of an irreducible representation can then be labeled by eigenvalues of H_i

$$H_i |p, q, \vec{\mu}\rangle = \mu_i |p, q, \vec{\mu}\rangle \quad (24)$$

In the case of fundamental representation of $SU(3)$, we have three basis states labeled by (apart from $p = 1, q = 0$)

$$\vec{\mu} = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}\right), \left(-\frac{1}{2}, \frac{\sqrt{3}}{6}\right), \left(0, -\frac{\sqrt{3}}{3}\right) \quad (25)$$

The vectors $\vec{\mu}$ are called the **weight vectors** of the algebra.

Now we can compose ‘raising’ and ‘lowering’ operators out of the other six generators of $su(3)$. Similar to $SU(2)$, the raising and lowering operators can be chosen to have only one off-diagonal element. Define

$$E_{\pm 1,0} = \frac{1}{\sqrt{2}} (T_1 \pm iT_2), E_{\pm 1/2, \pm \sqrt{3}/2} = \frac{1}{\sqrt{2}} (T_4 \pm iT_5), E_{\mp 1/2, \pm \sqrt{3}/2} = \frac{1}{\sqrt{2}} (T_6 \pm iT_7) \quad (26)$$

Their subscript comes from their commutation relation with T_3 and T_8 and is related to how they change the weight vector. For example

$$[T_3, E_{\pm 1,0}] = \frac{1}{\sqrt{2}} ([T_3, T_1] \pm i[T_3, T_2]) = (\pm 1)E_{\pm 1,0}, [T_8, E_{\pm 1,0}] = \frac{1}{\sqrt{2}} ([T_8, T_1] \pm i[T_8, T_2]) = 0E_{\pm 1,0} \quad (27)$$

Correspondingly, application of $E_{\pm 1,0}$ to a state $|\vec{\mu}\rangle$ maps it to state $|\vec{\mu} + (\pm 1, 0)\rangle$. The vectors

$$\vec{\alpha} = (\pm 1, 0), (\pm 1/2, \pm \sqrt{3}/2), (\mp 1/2, \pm \sqrt{3}/2) \quad (28)$$

are called **roots** of the algebra. Moreover, we can find

$$[E_{\vec{\alpha}}, E_{-\vec{\alpha}}] = \vec{\alpha} \cdot \vec{H} \quad (29)$$

which is a generalization of $SU(2)$ relation $[J_+, J_-] = 2J_z$. Note that while the weight vector is specific to each representation, the root vector is the same for all representations.

Now we can repeat the exercise of $SU(2)$ and try to find the set of weight vectors which make up a representation. In particular, for a particular p and q , we can start from a particular weight vector, change it using the raising and lowering operators. By calculating how the norm of the basis state changes under such mapping (a function of p , q , $\vec{\mu}$ and $\vec{\alpha}$), we can find the ‘highest weight vector’ and the ‘lowest weight vector’ and hence the dimension of the representation and the action of all generators in this basis. In this way, we can determine all irreducible representations of $SU(3)$. However, this calculation is too complicated and we are not going to do it explicitly.

Instead, we only mention here a few important representation of $SU(3)$. It turns out the dimension of a particular representation $D(p, q)$ is

$$d(p, q) = \frac{1}{2}(p+1)(q+1)(p+q+2). \quad (30)$$

1. $D(0, 0)$

The dimension of this representation is 1 and this is the trivial representation. Every generator is represented as 0 and every group element is represented as 1. Still this is a very important representation in physics and it is called the **singlet** representation.

2. $D(1, 0)$

This is the 3 dimensional representation given by the special unitary matrices. It is called the **fundamental** representation. We have discussed a lot about this representation above.

3. $D(1, 1)$

This representation is 8 dimensional. This is an important representation called the **adjoint** representation. The adjoint representation is a representation of a Lie group on the vector space of its Lie algebra. The $SU(3)$ group has 8 generators, therefore, its adjoint representation is 8 dimensional. The adjoint representation is obtained by interpreting the commutation relation

$$[\hat{T}_a, T_b] = if_{abc}T_c \quad (31)$$

as the action of $\hat{T}_a$ on the basis T_b of the Lie algebra, mapping them to different linear combinations of the basis. For example, because

$$[T_1, T_2] = iT_3, [T_1, T_3] = -iT_2, [T_1, T_4] = i\frac{1}{2}T_7, [T_1, T_7] = -i\frac{1}{2}T_4, [T_1, T_5] = -i\frac{1}{2}T_6, [T_1, T_6] = i\frac{1}{2}T_5 \quad (32)$$

Therefore, T_1 is represented by the 8×8 matrix

$$T_1^{\text{adjoint}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (33)$$

We can see that if we change the basis of the Lie algebra to that of the Cartan operator and the raising and lowering operator, the matrix corresponding to T_3 and T_8 becomes diagonal with the diagonal vector being the root of the algebra.